Urban Data: From Pain Points to Learning Pathways

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PIA Urban Data Network - WA
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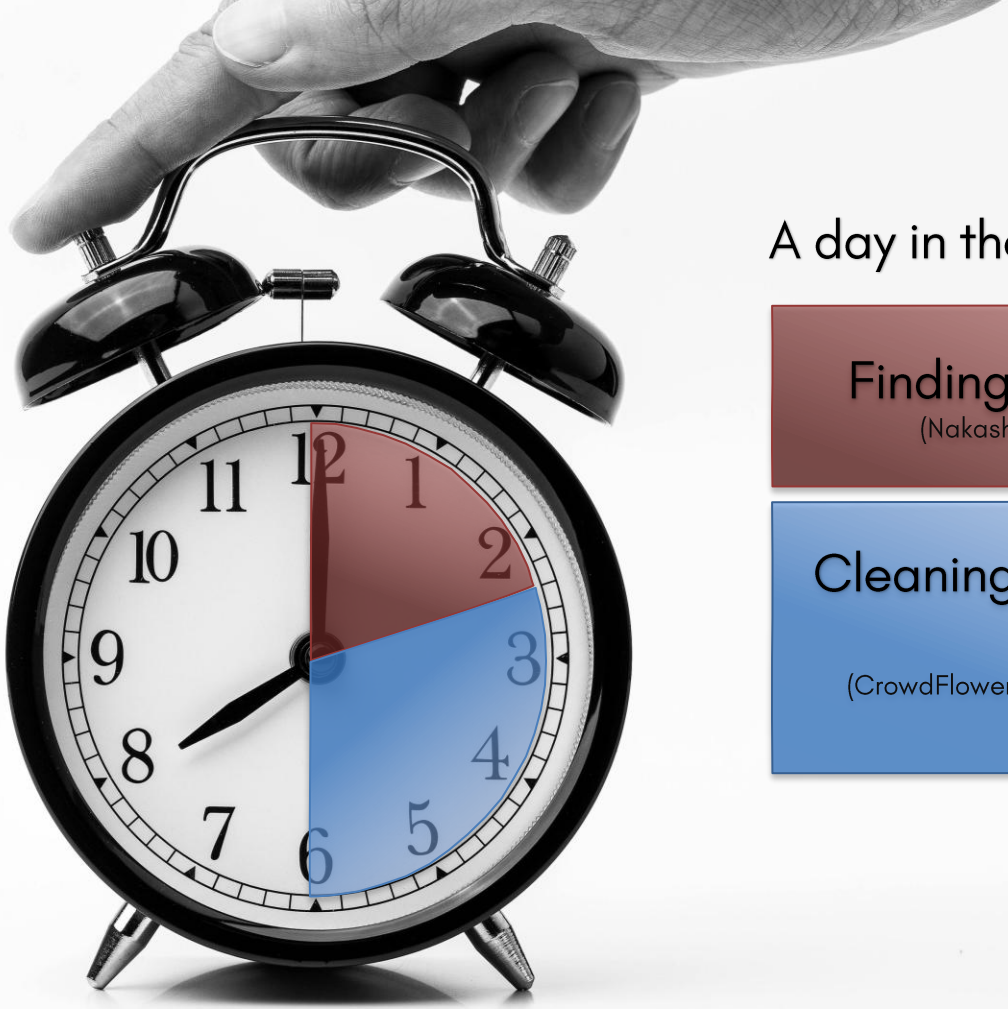


A day in the life of a data analyst…

Finding data ~ 2.4 hours day
(Nakash & Brouhnik 2024, Atlassian 2025)

Cleaning and organising data ~ 3.6 hours day
(CrowdFlower 2020, Kaggle 2019-, Anaconda 2020-)

Part 1: Pain Points

Why things take so long…

Finding urban data

- Fragmented sources & data silos
- Inconsistent formats & metadata
- Accessibility & usability challenges
- Temporal & spatial inconsistencies

Analysing urban data

- Data quality & reliability
- Lack of interoperability & linkage
- Technological barriers
- Skills gap in analytical techniques

Foundational Skills

1. **Data Literacy** – *read, understand & communicate with data*

2. **Critical Thinking** – *questioning the source, purpose and usefulness of data (in addressing the 'problem')*

Core competencies

Build Skills & Learn Tools

- Learning is iterative and practical application helps reinforce theory – *find a project*
- Blend technical, analytical, and domain-specific knowledge

Technical Skills:

- Geographic Information Systems (GIS)
- Programming for data analysis
- Data visualisation tools

Universal Skill:

- Writing and presentation for clear communication – adopt style guides (eg. https://www.stylemanual.gov.au)

Part 2: Learning Pathways

Helpful Resources

Learning resources

THE CARPENTRIES

DATA CARPENTRY

library carpentry

software carpentry

Australian Government Style Manual

Office for National Statistics Service manual

Medium

stackoverflow

MIT Open Learning

edX

MOOC

meetup

YouTube AU

Playlists (curated syllabus)

Data sources

Australian Bureau of Statistics Stat Data Explorer Beta

Australian Bureau of Statistics TableBuilder

Australian Government Data.gov.au beta

dataWA

NSW GOVERNMENT Data.NSW

VIC.GOV.AU DATAVIC

Data SA South Australian Government Data Directory

ACT Government ACTmapi

Queensland Government Open Data Portal

Queensland Government Queensland Government Statistician's Office

Google Dataset Search https://datasetsearch.research.google.com/

Australian Government Jobs and Skills Australia

An Australian Government Initiative DIGITAL ATLAS OF AUSTRALIA

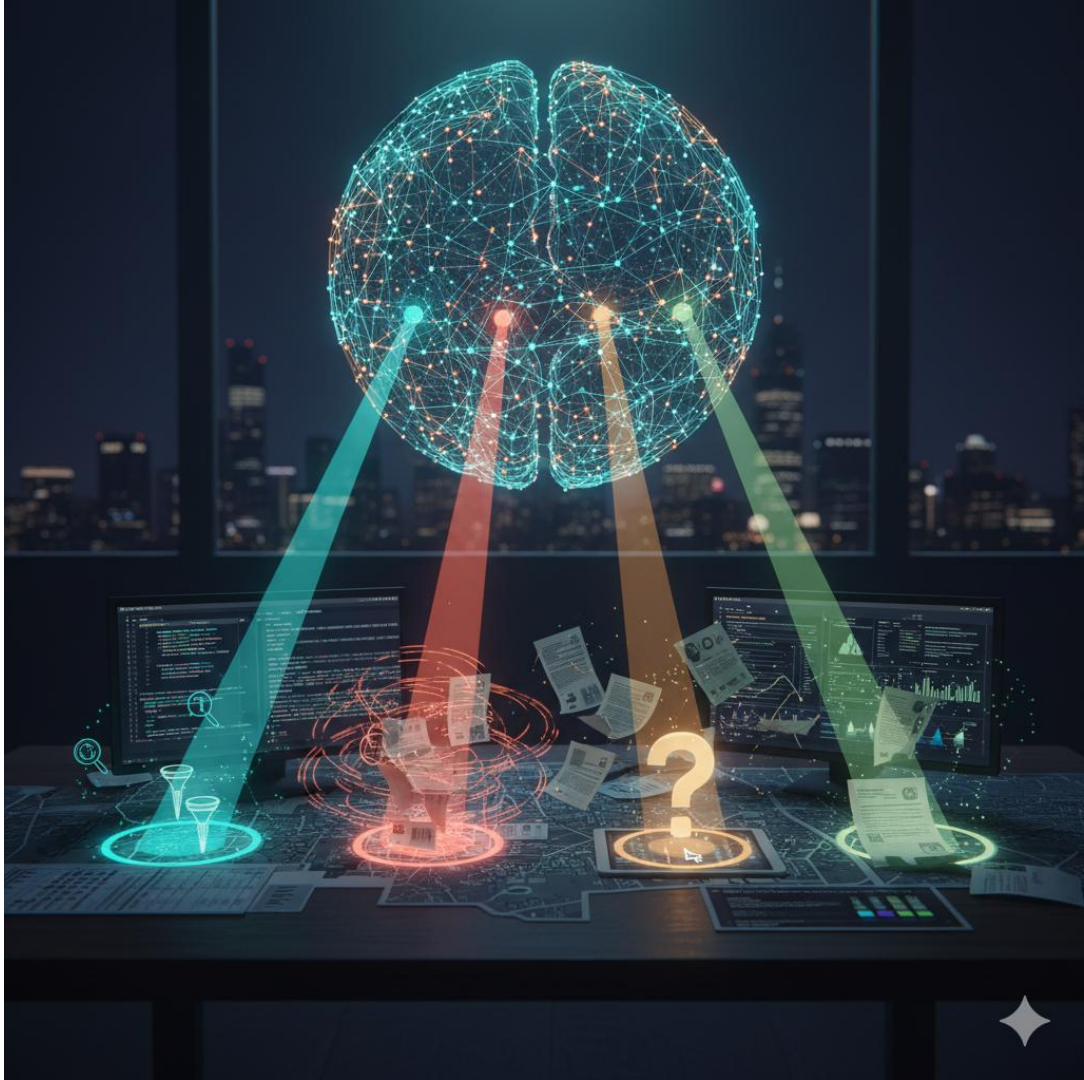
Australian Government Centre for Population

Australian Government Geoscience Australia

Australian Government Department of Employment and Workplace Relations

AI and Agents

- AI can find, filter, summarise and document
- AI uses *predictive* modelling and natural language processing – outcomes are probabilistic – a very different approach to purely algorithmic computations
- AI can't be trusted to analyse data directly because it uses probabilistic language models, not pure mathematic algorithms. However, AI agents *can* build algorithms (using R, Python, SQL) that *can* be independently validated



Summary

Find a project – something that has a value to your work/interests now

Foundations matter, but don't use it as a reason not to start

Build a skill set – balance theory and practice, learn to read code, even if you don't intend to write it. The language does not matter (if in doubt, choose Python)

Toolkit - include knowledge management tools, online tools, and a good text editor is your best friend

AI - a learning & research tool, but agents are not always reliable.

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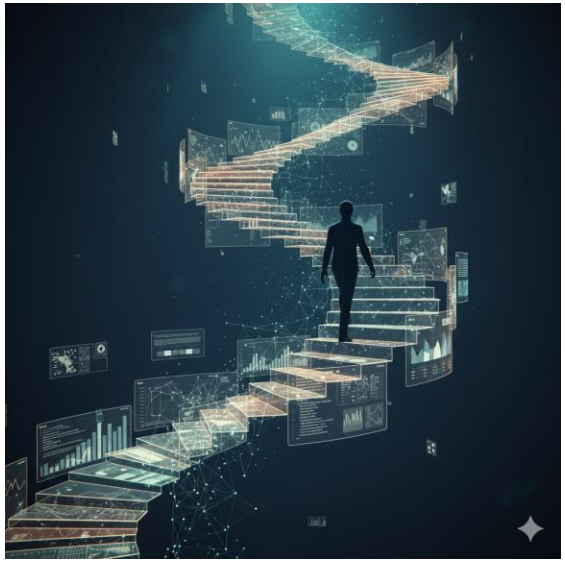

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